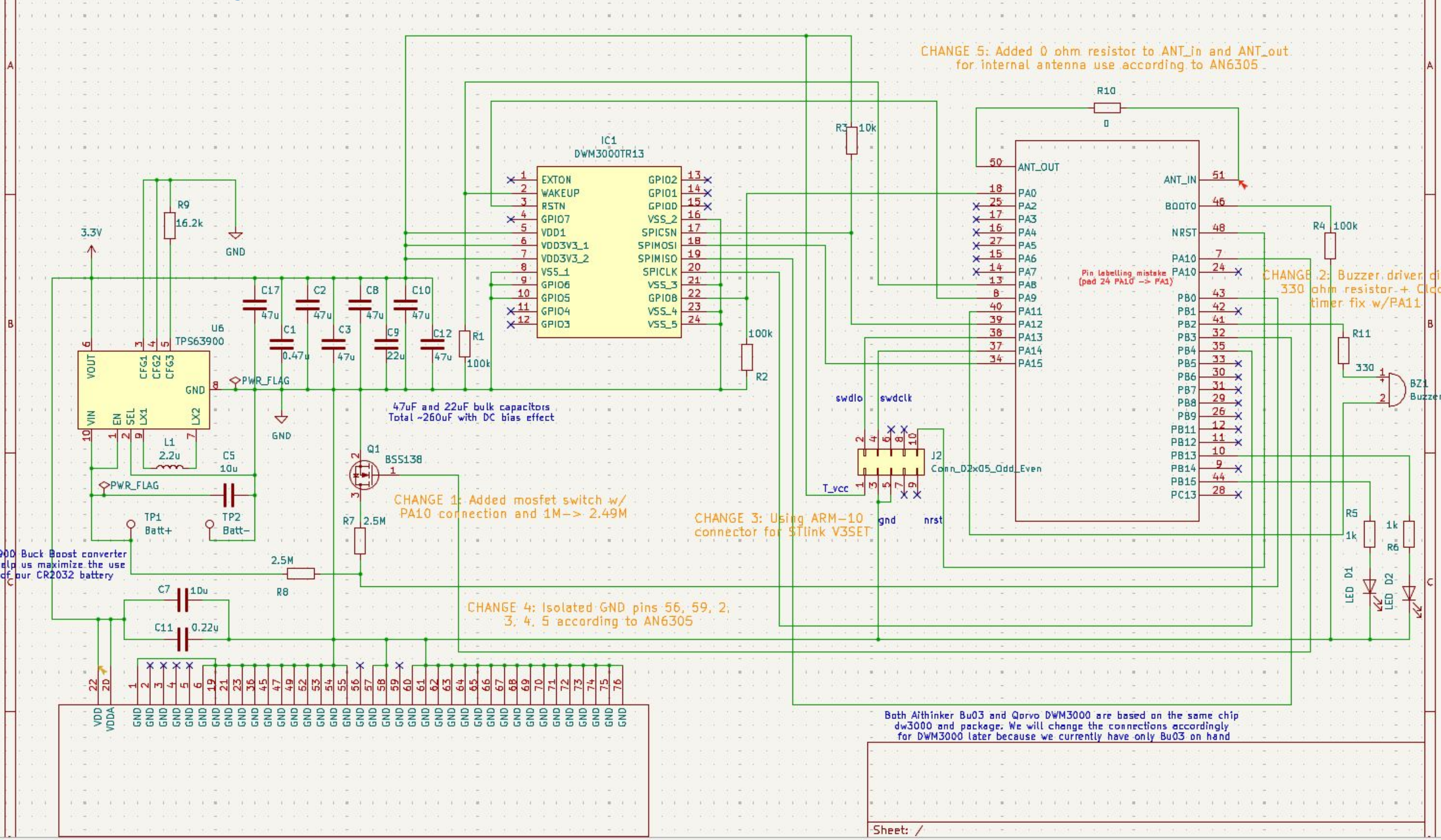
Compact Tag w/ STM32WBA55

- -Battery Life Optimization
- -Communication Modes
- -Productization Plan

Current System schematic



DWM3000 + MLCC Ideal Parameters & Lifespan

SYSTEM PARAMETERS

Total Usable Battery:
195.5 mAh
(230 mAh CR2032 @ 85% Derating)

Active Pulse Current:
63.0 mA (3 ms duration)

Deep Sleep Floor (**3.26 µA** total):

- DWM3000: 0.26 µA
- STM32WBA55: 1.50 µA
- TPS63900: 0.075 µA
- MLCC Leakage: ~ 0 µA

SCENARIO A: 1 HZ

1 Pulse per Second (86,400/Day)

Active Drain / Day:
3.841 mAh

Sleep Drain / Day:
0.163 mAh

Total Daily Drain:
4.004 mAh/day

ESTIMATED LIFESPAN

~48.8 Days

SCENARIO B: 3 HZ

3 Pulses per Second (259,200/Day)

Active Drain / Day:
11.524 mAh

Sleep Drain / Day:
0.162 mAh

Total Daily Drain:
11.686 mAh/day

ESTIMATED LIFESPAN

~16.7 Days

Full tag system Realistic Parameters & Lifespan

SYSTEM PARAMETERS

Total Usable Battery:
195.5 mAh
(230 mAh CR2032 @ 85% Derating)

Active Pulse Current:
72.5 mA (3 ms duration, Buck-Boost efficiency = 85% at Vin = 2.9V)

Deep Sleep Floor (**8.76 μA** total):

- DWM3000: 0.26 μA
- STM32WBA55: 6.50 μA
- TPS 63900: 0.075 μA
- MLCC Leakage: ~ 2 μA

At 85% efficiency, Deep Sleep Floor current pull from battery = 11.76 μA

Total Sleep floor = 11.8 μA

SCENARIO A: 1 HZ

1 Pulse per Second (86,400/Day)

Active Drain / Day:
5.220 mAh

Sleep Drain / Day:
0.282 mAh

Total Daily Drain:
5.502 mAh/day

ESTIMATED LIFESPAN

~35.5 Days

Assuming UWB usage 15 mins/ week

~13.5 Months

SCENARIO B: 3 HZ

3 Pulses per Second (259,200/Day)

Active Drain / Day:
15.660 mAh

Sleep Drain / Day:
0.281 mAh

Total Daily Drain:
15.941 mAh/day

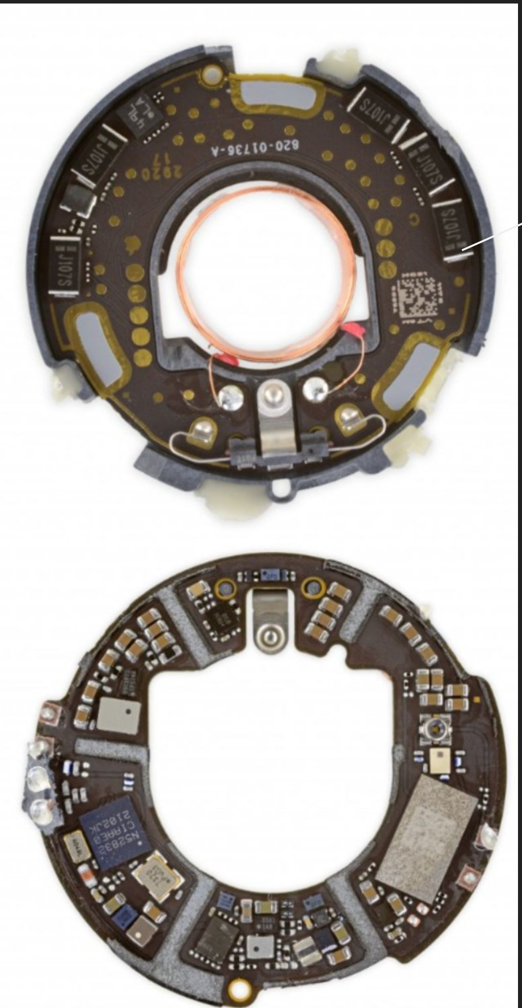
ESTIMATED LIFESPAN

~12.3 Days

Assuming UWB usage 15 mins/ week

~13 Months

Apple Airtag Power delivery inspiration



Capacitance rating (107) : 100 UF
Voltage Rating (J) : 6.3V
Tolerance (S) : +50% , -20%
Type : Polymer Tantalum

We can see 5 of these so net total capacitance is around 500 uF

Our system uses following Bulk MLCCs:
6 x 47uF(25V) = 282uF
1 x 22uF(25V) = 22uF
Net Bulk capacitance = 304uF

The usage of MLCCs allows for no leakage capacitance and gives us a better UWB lifespan.

Major Changes Needed for Productization

Change 1: Integration with BLE(Bluetooth Low Energy) technology

- We utilize the UWB module with another BLE setup (like the one we saw in our visit or BlueNRG-M2 or STM32WBA5MMG+ANT3216A063R2400A)

https://www.digikey.in/en/products/detail/yageo/ANT3216A063R2400A/311-1542-1-ND/3927225

https://www.st.com/en/wireless-connectivity/bluenrg-m2.html

- We can use the BLE setup 99% (UWB deep sleep, 142 nA) of the time to conserve battery and get to proximity of the tag without bothering with UWB unless we get close to the tag (<20m) .

Change 2: Double-Sided Two-Way Ranging (DS-TWR) for UWB communication

Changing our older single-sided ranging system for DS-TWR:

- **Stage-1** (Tag Tx , Anchor Rx)- Tag transmits signal for the initiation of the distance + direction measurement.
- **Stage-2** (Tag Rx , Anchor Tx)- Anchor transmits a signal and also starts counter for ToF then tag receives.
- **Stage-3** (Tag Tx , Anchor Rx)- Tag transmits signal while the Anchor receives and stops ToF counter , to find distance + direction.

Table 4 Power consumption table

Mode	Min.	Typical value	Max.	Unit
CH5 transmitting at 0.85Mbps	-	18.77	-	mA
CH5 transmitting at 6.81Mbps	-	16.25	-	mA
CH9 transmitting at 0.85Mbps	-	26.43	-	mA
CH9 transmitting at 6.81Mbps	-	24.89	-	mA
CH5 receives at 0.85Mbps	-	44.17	-	mA
CH5 receives at 6.81Mbps	-	44.14	-	mA
CH9 receives at 0.85Mbps	-	53.97	-	mA
CH9 receives at 6.81Mbps	-	53.74	-	mA
Instantaneous start-up current	-	176	-	mA
Deep sleep	-	142	-	nA

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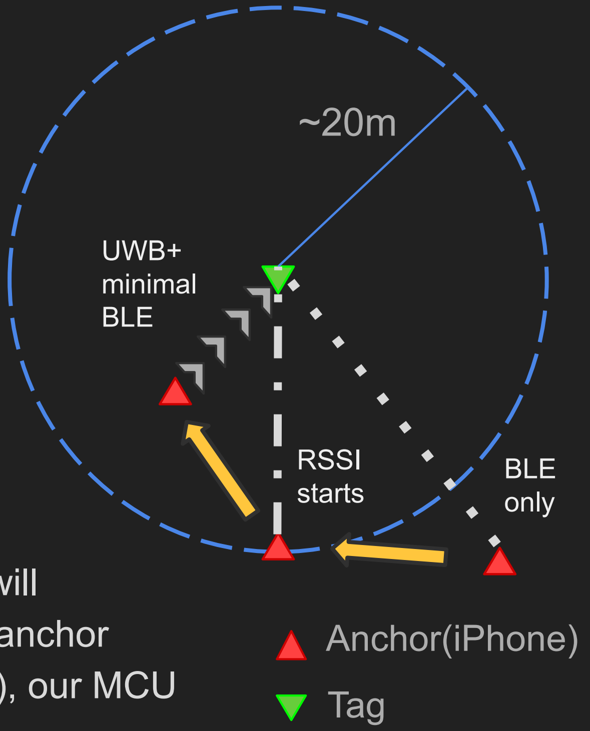
Change-1 detailed

- For most of the battery life ,our UWB subsystem will be **virtually nonexistent** and we will only use BLE technology with Find My app (on iOS w/ UWB support) or Find My Device(by Google,No UWB support).
- Our Tag BLE subsystem will act **like a beacon** that will broadcast its ID every 2s to every nearby **device in its own ecosystem** (say iOS) .
- The device will have a GPS location that will upload the **Tag ID** to **Apple cloud** which will get picked up by our own Anchor (iPhone) that will give us a rough location as to where our tag is.
- Once we get to a reasonable proximity (about 20m), our BLE subsystem will start using its **RSSI indicator** to find rough distance between the tag and anchor (accuracy in meters) and when the strength is adequate (around -65 dBm), our MCU will get our UWB subsystem **out of Deep-Sleep** and into active mode.
- Then we can use our iPhone (with U2/U1 chip) to track the direction and distance to our **DW3000 chip based** UWB module/setup without our **Change-2** tracking method. **Qorvo Nearby Interaction** or Apple's Find My app can handle this part well.



Getting Access to Apple's Find My

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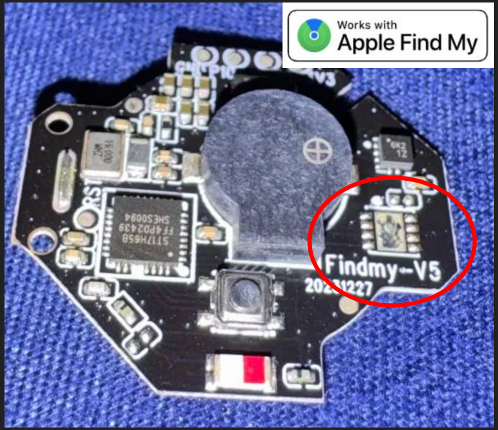
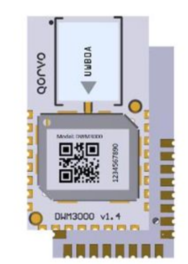
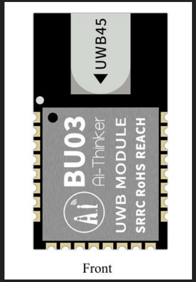
https://mfi.apple.com/en/how-it-works.html

Portronics World Tag







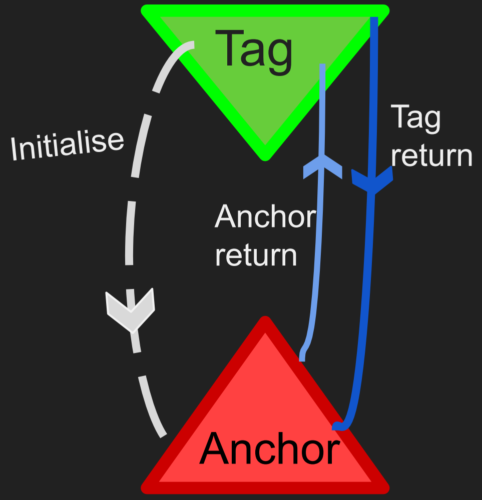
Apple allows third party devices to enter their iOS ecosystem if they comply with certain guidelines* using the MFi program and have suitable BLE technology in your circuit.

A temporary work-around we can use is a masked ID to utilize Apple's Find My network for BLE purposes and Qorvo's nearby interaction for UWB tracking.

https://github.com/seemoo-lab/openhaystack

Dw3000 based modules that we plan to use are compliant with IEEE 802.15.4z standard of UWB communication which is also found in UWB subsystems of higher-end android phones.

Change-2 detailed

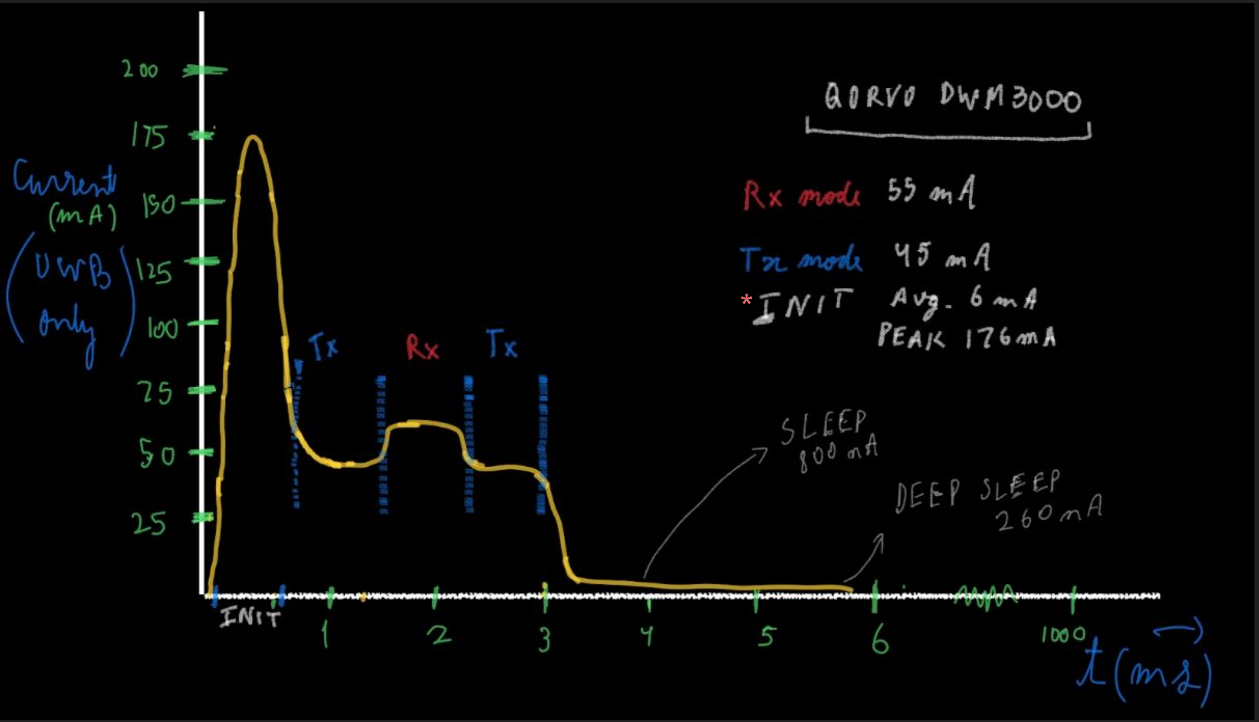


The industry standard method to use UWB communication in unknown terrain is DS-TWR in which we have a 3-signal broadcast (Initialise, Anchor return and Tag return in order)

1. **Initialize :** Tag in Tx mode sends a signal to Anchor to basically inform the anchor that the tag is reachable for distance calculation, just like t_ready signal sent by slave in AXI.
2. **Anchor return :** Anchor sends the starter signal for ToF calculation and gets into Rx mode for the Tag return signal.
3. **Tag return :** Tag sends the signal back to anchor to end the ToF on anchor side and our iPhone anchor is capable of finding the distance and direction from the ToF alone.

TAG

DS-TWR current pull pattern



The diagram gives a rough idea to what our current consumption by the UWB module will be during its active phase in the CH-5 mode.

We would be able to keep the dwm3000 in deep sleep during 99.5% of the time cycle (here 1s)

Since our cr2032 battery won't be enough to complete these high current requirements ,we would need to mostly rely on reservoir capacitors to power dwm3000 in the active duration (planned to be around 3ms)

*INIT stands for UWB initialization here and not the Initialize signal